LIVENOTE: REAL-TIME AI MEETING TRANSCRIPTION, SUMMARIZATION & ACTION ITEM EXTRACTION

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THE JOURNEY

🔍 The Problem:

- Meetings generate key Decisions and Tasks, but outcomes get lost in messy notes or long recordings
- Action items often lack clear Owner & Deadlines, causing missed follow-ups and low accountability
- Existing tools usually summarize after the meeting and require heavy manual cleanup
- **Impact**: Wasted time, Duplicated work, and Unreliable follow-through.

💡 Motivation:

- Meetings are one of the most common and least structured work artifacts
- A near real-time assistant reduces post-meeting overhead and makes accountability immediate
- **Goal**: To make meeting documentation usable during the meeting, not hours later

🤖 AI Techniques Used:

- ASR (Speech→Text): Whisper (Transformer-based speech model)
- Speaker diarization: Separate diarization model and Alignment to transcripts (Who spoke when)
- LLM Intelligence Core: Incremental summarization + Structured JSON extraction
  - Action items (Task, owner, Deadline, Priority)
  - Decisions
- Trust layer: Grounding and Rule checks (Evidence spans, Speaker validity, Deadline parsing)

🏆 Success Criteria:

- LiveNote produces Incremental meeting summaries and Action items every 2–3 minutes
- Action items are grounded (traceable to transcript evidence) and editable
- Demonstrates measurable performance: WER for transcription, ROUGE for summaries, and action extraction accuracy on labeled subset
- Runs in near real-time with stable latency per chunk (prototype deployment path validated)

DATA STRESS TEST

RAW DATA INGEST

a. AMI Dataset Version-1: knkarthick/AMI

```
from datasets import load_dataset

ds1 = load_dataset("knkarthick/AMI", token="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")
```

```
Using the latest cached version of the dataset since knkarthick/AMI couldn't be found on the Hugging Face Hub
Found the latest cached dataset configuration 'default' at /Users/satwik/.cache/huggingface/datasets/knkarthick___ami/default/0.0.0/51ee8e2
2888b3aafb4a2601796c76c8fd750ebfd (last modified on Sun Feb 15 18:38:31 2026).
```

```
#Dataset Structure
print(ds1)
```

```
DatasetDict({
    train: Dataset({
        features: ['id', 'dialogue', 'summary'],
        num_rows: 209
    })
    validation: Dataset({
        features: ['id', 'dialogue', 'summary'],
        num_rows: 42
    })
    test: Dataset({
        features: ['id', 'dialogue', 'summary'],
        num_rows: 28
    })
})
```

```
#Column Names
print(ds1["train"].column_names)
```

```
['id', 'dialogue', 'summary']
```

b. AMI Dataset Version-2: edinburghcstr/ami (ihm)

```
from datasets import load_dataset

ds2 = load_dataset("edinburghcstr/ami", "ihm", token="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")
```

```
Using the latest cached version of the dataset since edinburghcstr/ami couldn't be found on the Hugging Face Hub
Found the latest cached dataset configuration 'ihm' at /Users/satwik/.cache/huggingface/datasets/edinburghcstr___ami/ihm/0.0.0/46f28f2503e2
ec48f8867a84eef356c70476beab (last modified on Mon Feb 16 20:34:46 2026).
Loading dataset shards: 100% ▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰ 36/36 [00:05<00:00, 7.16it/s]
```

```
ds2
```

```
DatasetDict({
    train: Dataset({
        features: ['meeting_id', 'audio_id', 'text', 'audio', 'begin_time', 'end_time', 'microphone_id', 'speaker_id'],
        num_rows: 108502
    })
    validation: Dataset({
        features: ['meeting_id', 'audio_id', 'text', 'audio', 'begin_time', 'end_time', 'microphone_id', 'speaker_id'],
        num_rows: 13098
    })
    test: Dataset({
        features: ['meeting_id', 'audio_id', 'text', 'audio', 'begin_time', 'end_time', 'microphone_id', 'speaker_id'],
        num_rows: 12643
    })
})
```

```
ds2["train"].features
```

```
{'meeting_id': Value('string'),
 'audio_id': Value('string'),
 'text': Value('string'),
 'audio': Audio(sampling_rate=16000, decode=True, stream_index=None),
 'begin_time': Value('float32'),
 'end_time': Value('float32'),
 'microphone_id': Value('string'),
 'speaker_id': Value('string')}
```

c. ICSI Dataset argmaxinc/icsi-meetings

```
from datasets import load_dataset

ds_icsi = load_dataset("argmaxinc/icsi-meetings", token="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")
```

```
Using the latest cached version of the dataset since argmaxinc/icsi-meetings couldn't be found on the Hugging Face Hub
Found the latest cached dataset configuration 'default' at /Users/satwik/.cache/huggingface/datasets/argmaxinc___icsi-meetings/default/0.0.
0/a6c0202a27845d86cd5c0f53c299725fb219fdad (last modified on Wed Feb 18 11:14:06 2026).
```

```
# Dataset structure
ds_icsi

# Split names
print(ds_icsi.keys())

# Number of samples per split
for split in ds_icsi:
    print(split, len(ds_icsi[split]))
```

```
dict_keys(['test'])
test 75
```

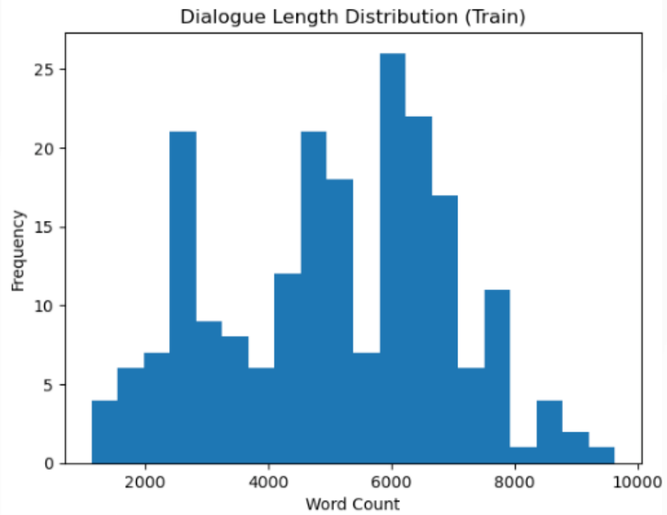
```
# Column names
print(ds_icsi["test"].column_names)

# Feature types
print(ds_icsi["test"].features)
```

```
['audio', 'timestamps_start', 'timestamps_end', 'speakers']
{'audio': Audio(sampling_rate=16000, decode=True, stream_index=None), 'timestamps_start': List(Value('float64')), 'timestamps_end': List(Va
lue('float64')), 'speakers': List(Value('string'))}
```

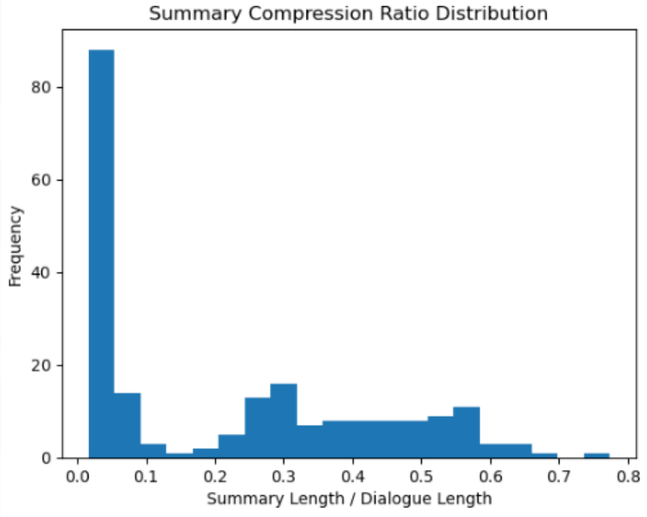
DATA STRESS TEST

EXPLORATORY DATA ANALYSIS (EDA)

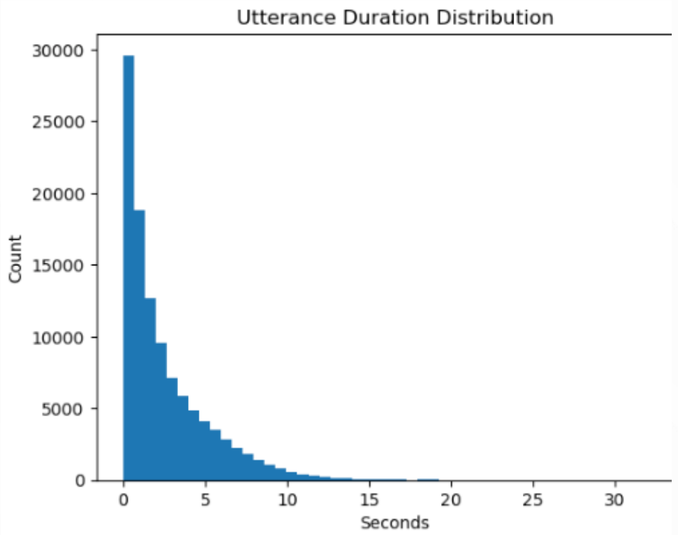
a. Dialogue Length Distribution (knkarthick/AMI)



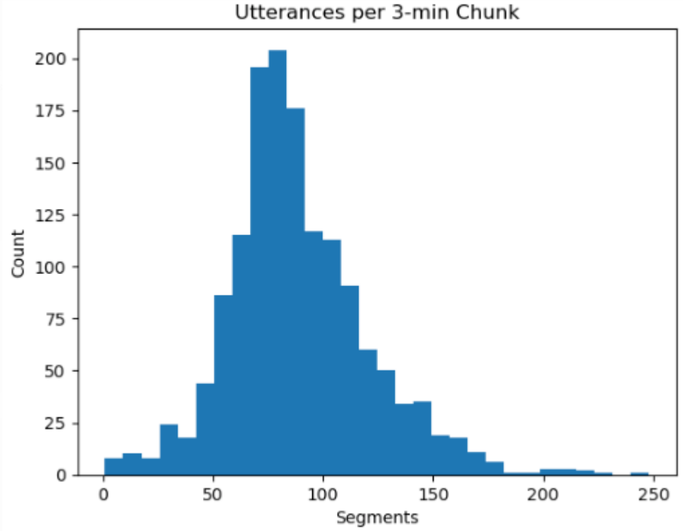
b. Compression Ratio Distribution (knkarthick/AMI)



c. Utterance Duration Distribution (edinburghcstr/ami ihm)



d. Utterances per 3-min Chunk (ICSI)



MODELING & BASELINE COMPARISON

Baseline vs Transformer: Summarization Performance (AMI, n=28)

Naive Baseline : Lead-3 Extractive

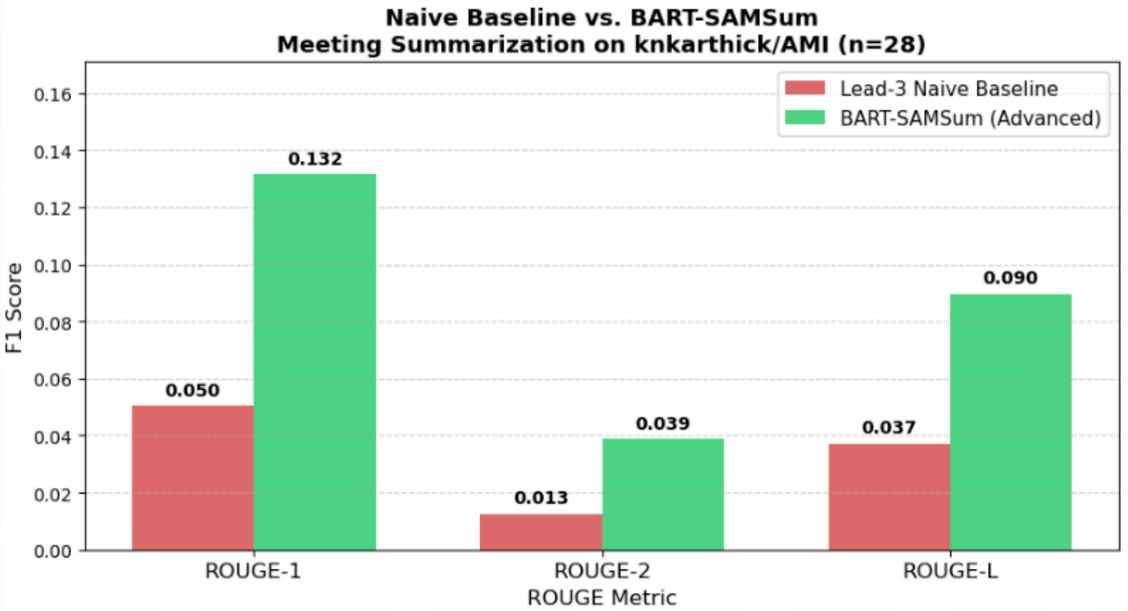
- **Method:** First 3 sentences of the dialogue
- **Purpose:** To establish a performance floor (Zero ML/training)

Advanced Model : Transformer Summarizer

- **Model:** philschmid/bart-large-cnn-samsum (BART fine-tuned for dialogue summaries)
- **Setup:** CPU inference, truncated to 3000 chars due to 1024-token input limit

Results

(Mean ROUGE-F1)

Method	ROUGE-1	ROUGE-2	ROUGE-L
Lead-3 (Naive)	0.0503	0.0126	0.0371
BART-SAMSum (Advanced)	0.1315	0.0387	0.0896
Δ Improvement	+0.0812 (+161%)	+0.0261 (+207%)	+0.0525 (+142%)

Key Finding: BART-SAMSum shows significant improvement over the baseline Lead-3 approach, with the largest gain in ROUGE-2 (+207%), indicating better bigram overlap in generated summaries.